Experiment – 2

Aim: Study of Sensors.

Theory:

A sensor is a device that detects and responds to some type of input from the physical environment. The input can be light, heat, motion, moisture, pressure or any number of other environmental phenomena.

Types of Sensors

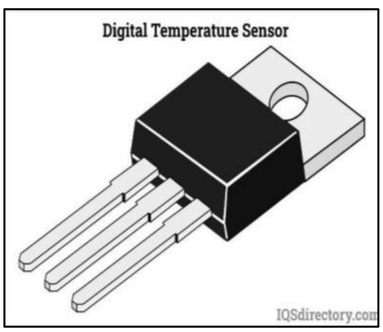
1. **Temperature Sensors**

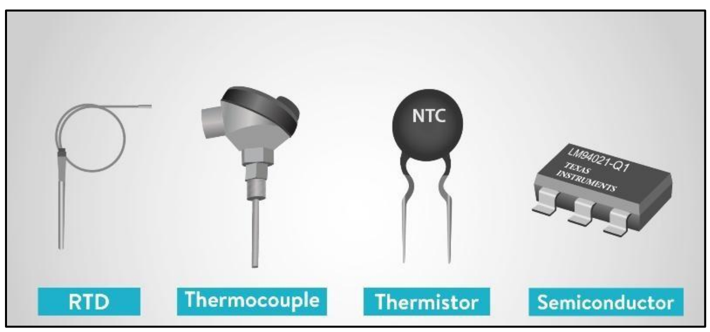
   Function: Measure temperature variations by detecting changes in electrical properties.

   Examples:

   - Thermocouples: Generate voltage based on temperature differences, commonly used in industrial applications.
   - RTDs (Resistance Temperature Detectors): Change resistance proportionally to temperature, offering high accuracy and stability.
   - Infrared (IR) Sensors: Measure temperature remotely by detecting infrared radiation, useful in medical and industrial applications.

   Applications: HVAC systems, weather monitoring, medical thermometers, industrial temperature control.





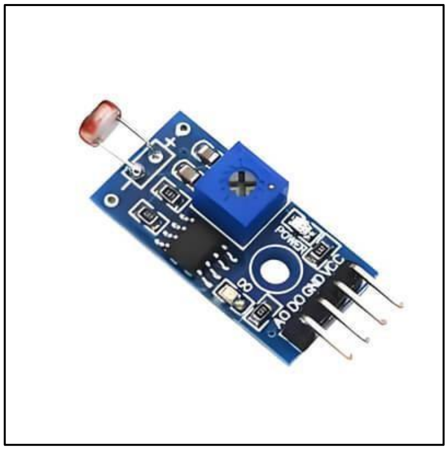
2. **Light Sensors**

Function: Detect and measure the intensity of light.

Examples:

- Photodiodes: Convert light into an electrical current, commonly used in optical communication and safety systems.
- LDRs (Light Dependent Resistors): Change resistance with light intensity, used in automatic lighting and display adjustments.

Applications: Automatic streetlights, smartphone brightness control, solar panels, security systems.



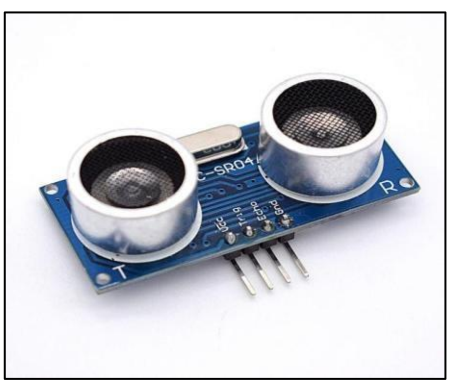
3. **Ultrasonic Sensors**

Function: Use high-frequency sound waves to measure distance or detect objects.

Examples:

- HC-SR04 Sensor: Uses ultrasonic waves to determine distances with high precision, widely used in robotics.

Applications: Robotics, liquid level sensing, vehicle blind-spot detection, parking assistance, obstacle avoidance systems.



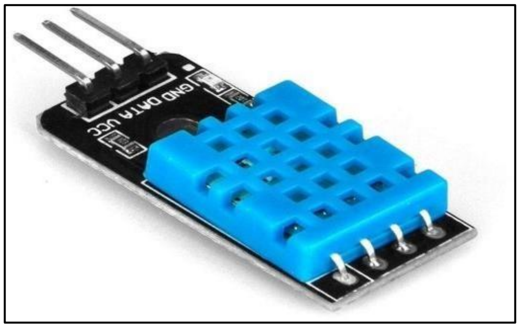
4. Humidity Sensors

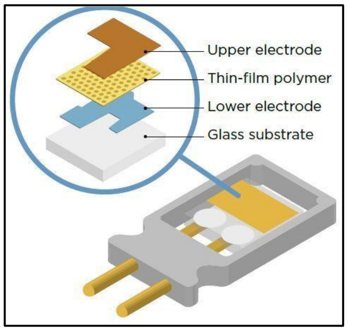
Function: Measure moisture content in the air to ensure optimal environmental conditions.

Examples:

- Capacitive Sensors: Detect changes in capacitance due to humidity variations.
- Resistive Sensors: Measure changes in resistance caused by moisture levels.
- Thermal Sensors: Use heat conductivity variations to determine humidity.

Applications: Weather monitoring, HVAC systems, food storage, agriculture, industrial processes.





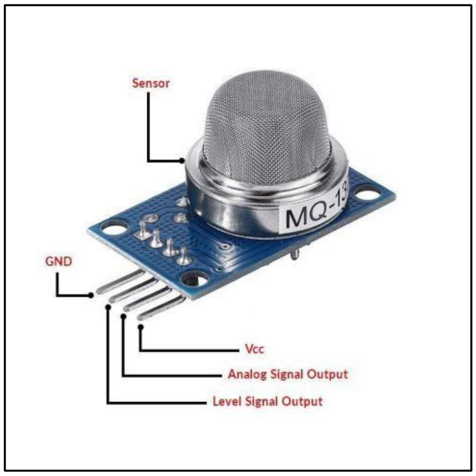
5. Gas Sensors

Function: Detect and measure the presence of gases in the environment for safety and monitoring.

Examples:

- MQ Series Sensors (MQ-2, MQ-7): Detect gases like methane, carbon monoxide, and smoke.
- Electrochemical Sensors: Measure gas concentration through chemical reactions that generate electrical signals.

Applications: Air quality monitoring, smoke detectors, industrial safety systems, automotive emission control.



Conclusion:

Sensors are fundamental to modern technology, enabling real-time monitoring and smart automation in various industries. Their integration with IoT and AI enhances precision and efficiency, driving innovation in industrial, medical, and consumer applications. Ongoing advancements continue to improve sensor performance, making systems more intelligent, responsive, and efficient.